

**Data Technologies —**  
**Stat 5080, Section 001 & Stat 6080, Section 001**  
**Fall 2026 (2 Credits)**

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[https://symanzik.github.io/teaching/2026\\_stat5080/stat5080.html](https://symanzik.github.io/teaching/2026_stat5080/stat5080.html)

Office Hours: Tuesday (T) & Thursday (H) 1:30pm – 2:30pm in RWST 416,  
Friday (F) 9:00am – 10:00am (via Zoom), and by appointment.

Classes & Rooms:  
TH 12:00noon – 1:15pm, T 9/15 – H 11/19, 2026 (tentatively): HPER 111E (face-to-face).

Teaching Assistant: None.

Please visit the course Web page listed above for emergency announcements, e.g., when Canvas is unavailable. Otherwise, visit Canvas frequently for lecture notes, data sets, R code, etc. — in particular if you miss our face-to-face lectures for any reason. All (additional and updated) materials, announcements, discussions, recordings, etc. from Canvas are part of the course materials. Not seeing one of these in time does not serve as an excuse for not getting point deductions for the course. Deadlines may change or unexpected new regulations and requirements may occur. It is your responsibility to make sure to receive all announcements in time.

Detailed Class Schedule:

For a 2-credit course, we need 20 lectures/lecture days (in contrast to 29 or 30 lectures/lecture days for a 3-credit course). Those days are marked as “Lecture 01” to “Lecture 20” in the overview below:

| Week | Tuesday           | Thursday          |
|------|-------------------|-------------------|
| 1    | 9/1 No class      | 9/3: No class     |
| 2    | 9/8: No class     | 9/10: No class    |
| 3    | 9/15: Lecture 01  | 9/17: Lecture 02  |
| 4    | 9/22: Lecture 03  | 9/24: Lecture 04  |
| 5    | 9/29: Lecture 05  | 10/1: Lecture 06  |
| 6    | 10/6: Lecture 07  | 10/8: Lecture 08  |
| 7    | 10/13: Lecture 09 | 10/15: Lecture 10 |
| 8    | 10/20: Lecture 11 | 10/22: Lecture 12 |
| 9    | 10/27: Lecture 13 | 10/29: Lecture 14 |
| 10   | 11/3: Lecture 15  | 11/5: Lecture 16  |
| 11   | 11/10: Lecture 17 | 11/12: Lecture 18 |
| 12   | 11/17: Lecture 19 | 11/19: Lecture 20 |
| 13   | 11/24: Backup     | 11/26: No class   |
| 14   | 12/1: Backup      | 12/3: Backup      |
| 15   | 12/8: No class    | 12/10: No class   |

Note: “No class” means guaranteed no class that day. I have marked the dates in the three last weeks of the semester as “Backup”, e.g., in case we miss lectures because of weather conditions, sickness, or any other reason. If nothing goes wrong, our tentative last lecture date will be on H 11/19/2026. Backup lectures (if any) will be held via Zoom (and not face-to-face). There is no need to stay in Logan after H 11/19/2026 for a backup lecture in this course.

### Course Objectives:

Note that Andreas Buja, the Liem Sioe Liong/First Pacific Company Professor in the Statistics Department, The Wharton School, at the University of Pennsylvania in Philadelphia, USA, already stated in a 2006 interview: *“I think in education we still have some ways to go to find a balance of things that we want to teach students. There is the traditional curriculum that gives Ph.D. students a solid foundation in theory, but then they also should acquire computational skills, they should become good applied statisticians who have good sense, good data sense. Many of these things are really hard to teach. You can teach them details, but ultimately they have to pick up the high level of thinking, the creative way of thinking, on their own or by being thrown like fish into the water, either they swim or they don’t. That is hard to teach. Something that I see lacking right now is, especially if you are interested in education with industry in mind, what you need out there in industry I think is not specifics of modeling, it is good data sense, it is data skills, data literacy. I think that was a term used by one of the earlier interviewees. Data literacy, in general, is the ability to get data and start doing something sensible, and that is of utmost importance. Part of that is that we cannot assume that other people are doing data cleaning for us. We have to do that ourselves. So here I see a gap actually in our education. I don’t think most statistics programs teach something like a scripting language and practice data cleaning, reshaping of data, basic tabulations, mild aggregations, getting subsamples, systematic ones and random ones, and so on. These are very important activities, and we still need to get better at teaching them.”* (see Computational Statistics (2008) 23:177–184, <https://doi.org/10.1007/s00180-008-0113-0> for the full interview).

In the “Introduction to R” course, you have learned basic data skills and data literacy via R to achieve this goal. You were (likely) thrown like fish into the water (without any prior swimming course) — and you learned to swim! Now, we will extend these basic skills to

do something really meaningful with a variety of data sets. Eventually, this course will be one of your most valuable courses for a future career in research (assuming you are working with any kind of data) or in industry.

**Prerequisites:**

STAT 5050 (“Introduction to R”) with a C– or better. STAT 3000 (“Statistics for Scientists”) or STAT 5100 (“Modern Regression Methods”) with a C– or better. STAT 5550 (“Statistical Visualization I”) is recommended. Moreover, you should be familiar with a tool such as R Markdown, knitr, or Sweave that allows you to combine text, R code, graphics, and numerical results in high-quality documents. L<sup>A</sup>T<sub>E</sub>X is a plus, but is not formally required at the 5000 level, but it will be required at the 6000 level of this course.

Moreover, I expect that you still have “operational” knowledge from your STAT 3000 or STAT 5100 (or higher) course. “Operational” means that you still recall sufficient details from regression, ANOVA, hypothesis tests, etc. (it is not sufficient that you have taken such a course several years ago and have forgotten almost all details).

**IDEA Center Learning Objectives:**

**Objective 1)** Gaining a basic understanding of the subject (e.g., factual knowledge, methods, principles, generalizations, theories).

**Objective 4)** Developing specific skills, competencies, and points of view needed by professionals in the field most closely related to this course.

**Objective 13)** Learning appropriate methods for collecting, analyzing, and interpreting numerical information.

**Topics:** (subject to change)

1. Data.
2. Basics of simulation.
3. Representation of information.
4. Regular expressions.
5. HyperText Markup Language (HTML) and Web scraping.
6. Text data from Portable Document Format (pdf) files and via Optical Character Recognition (OCR).
7. The *Tidyverse* (R packages for Data Science).
8. Data bases and Structured Query Language (SQL).
9. Extensible Markup Language (XML).
10. Resampling/bootstrap/others (Outlook only).

We will work with real “messy” data that have not been preprocessed nor analyzed so far. These data will contain surprises — for you and for me. Do not expect that someone is going to give you the final answer or model. We jointly will have to work towards such an answer or model.

For MS and PhD students majoring in Statistics, it is important to learn L<sup>A</sup>T<sub>E</sub>X — from basic document preparation, over the inclusion of R graphics into your L<sup>A</sup>T<sub>E</sub>X documents to advanced topics such as Sweave (<https://stat.ethz.ch/R-manual/R-devel/library/utlils/doc/Sweave.pdf>), knitr (<https://yihui.org/knitr/>), and the L<sup>A</sup>T<sub>E</sub>X bibliography BibTeX (<http://www.bibtex.org/>). L<sup>A</sup>T<sub>E</sub>X is essential for graduate work (at the

MS and PhD level) and will be used for many theses, dissertations, and scientific publications. **Therefore, L<sup>A</sup>T<sub>E</sub>X will have to be used for all homeworks, projects, presentations, etc. at the 6000 level of this course.**

### **Course Format and Lecture Activity Points:**

The course will be offered in a face-to-face format, implying that you should attend most of the lectures in person. There will also be lecture recordings via Kaltura that will be posted in the “Media Gallery” in Canvas in the evenings of a lecture day. I will also open a Zoom link through Canvas on lecture days with planned in-class lecture activities so that you can (partially) participate in those activities and earn some credits for your remote participation. However, there will be limitations to what can be captured via the Kaltura recordings and what can be done via the Zoom link. In particular, the Zoom link will only be an audio link with no video and no view of my computer screen (as Kaltura and Zoom compete for the same devices on my laptop, in particular for the camera). So, try to attend the lectures in person whenever possible.

Under this setup, it will be possible to conduct activities during the lectures and award credit for participating in those activities, ideally in person, but alternatively via Zoom and/or the Canvas “Discussions”. You may be asked to answer some short questions during a lecture, do some programming and then submit your code into the Canvas “Discussions”, or fill out a worksheet on paper. Some of these activities are stand-alone while others build on each other during the course of a lecture (and thus the need to be able to participate in real time via Zoom). Some lectures may have extended sequences with around 10 (short) activities in a lecture.

Points earned through such activities count as Lecture Activity Points (LAPs). These activities typically will not be announced in advance. In fact, some activities may only be released in Canvas shortly before the start of a lecture. However, if there is no new activity visible in the Canvas “Discussions” at the start of a lecture and there is also no open Zoom link for that lecture, it is safe to assume that there will be no LAPs awarded in that lecture.

LAPs will be awarded on a done (answered) / not done (not answered) basis and not on correctness. This should encourage everyone to think about a question or try some coding without the pressure to do things immediately right when trying.

I cannot say in advance at this time how many lectures there will be where LAPs will be awarded, but I would estimate that there will be somewhere around  $12 \pm 5$  lectures with LAPs (out of the 20 lectures we will have this semester). In my previous courses, students could earn about 80 to 100 LAPs. I would predict that the total number of LAPs will be in a similar order of magnitude this semester. Some of the stand-alone activities may be answered prior to a lecture when posted in a Canvas “Discussion”. Some of the activities will remain open until 11:59pm on a lecture day and can still be answered after class. Extended sequences of activities typically cannot be answered prior to a lecture as they may require some of the lecture materials and answers to some of the earlier activities in such a sequence of activities.

**Points from LAPs will contribute to 10% of your course grade.** LAPs will translate as follows into course points:

| % of LAPs Completed | Course Points |
|---------------------|---------------|
| $[90, \infty)$      | 10            |
| $[80, 90)$          | 9             |
| $[70, 80)$          | 8             |
| $[60, 70)$          | 7             |
| $[50, 60)$          | 6             |
| $[0, 50)$           | 0             |

So, you can miss (for private reasons) some lectures where LAPs are provided and still get the full number of course points for LAPs. However, you should not miss too many of the lectures where a large number of LAPs will be provided. Also, make sure that you do not miss any LAPs later in the semester in case you already missed several LAPs early in the semester. Ideally, plan on attending all lectures in person.

In case of an excused absence, e.g., for medical reasons, family emergencies or funerals, court appointments, university-approved travel, etc., please provide some supporting information and your LAP score will be adjusted according to the number of lectures with LAPs you could attend. Private reasons such as travel, most family events (such as weddings), etc. do not count as an excused absence.

**Ideally, bring your laptop to all lectures so you can earn LAPs immediately during a lecture period.**

#### Homework Assignments:

There will be 3 HW assignments for this course, roughly one every three weeks. Each HW assignment will include a value (typically 20–100 points) that it will be scored out of. **HW assignments will contribute to 90% (for Stat 5080), respectively 60% (for Stat 6080), of your course grade.** The value of each HW assignment will be roughly proportional to its importance and the amount of work involved.

You will be allowed to discuss general approaches to questions on the HW assignments with other students, but each student must write and submit their own R code and comments. Any students caught sharing R code or other parts of their homework submissions will fail the class.

Unless otherwise stated on the HW assignment sheet, all homework assignments have to be submitted electronically via Canvas. HW assignments will be posted tentatively in weeks 6, 9, and 12 of the semester and will be tentatively due in weeks 9, 12, and 15 of the semester (see the *Detailed Class Schedule* above how these weeks approximately translate to dates). **You will have about 2 to 3 weeks after the last lecture to finalize and submit the last HW assignment.**

There will be no (in-class or take-home) quizzes, midterm exams, or final exams. Nevertheless, this will be a very challenging course that requires a lot of individual time to work on the assignments. Just attending classes will not be enough to pass this course! **In addition, you will have to do a lot of individual reading of textbooks, online documentation, and help pages, and search for available information on the web.**

The use of Artificial Intelligence (AI) tools such as ChatGPT, Co-Pilot, and others is permitted, but has to be acknowledged in your submissions.

#### No Excuse Needed Late Homework Submission Policy:

Each student has **3 tokens** for late homework submissions. These tokens can be used in

multiple ways, e.g., one token at a time for each of the 3 HWs or all 3 tokens can be used for a single late HW submission. You will need 1 token if your HW is 1 min – 24 hours late; you will need 2 tokens if your HW is > 24 hours – 48 hours late; and you will need all 3 tokens if your HW is > 48 hours – 72 hours late. Once you have used up all 3 tokens, late HWs will count as 0 points, even if such a HW is only one additional minute late. As an example, if you used 2 tokens early on, you will have only 1 token left and can submit your next HW only up to 24 hours late. If you submit that HW more than 24 hours late, it will count as 0 points. All times are based on the Canvas time stamp that is created when a HW gets submitted — and not on your local computer clock.

It is your responsibility to keep track of the number of tokens you have used. Record that information on your phone or check how you can extract the submission dates and times in Canvas on your side. I will not be able to tell you shortly before the next submission deadline how many tokens you have left. In fact, I only plan to use that information when I determine final course grades. You may therefore get assigned a score > 0 points for a certain HW submission initially, but that score may get adjusted to 0 points if it turns out that this was for a late submission and you had already used up all your tokens.

You will not get any credit for any unused tokens, so you could use the remaining ones for proofreading, checking that all figures and R code are included, etc., in particular for the later HWs — and if you still have tokens left at that time!

As the name indicates, there is no excuse needed when you use a token, be it for a minor sickness, personal travel due to family events, or just needing more time as you couldn't finish that HW by the deadline. If you are officially absent from USU when a HW is due, e.g., due to travel with a USU sports team or band, attending a conference, doing fieldwork, etc., I need an official statement from a coach or supervisor — and no token will be used. However, we then need to determine on a case-by-case basis when that particular HW will be due for you — and when you would have to use additional token(s) to further extend your extended deadline.

You can resubmit your HW as often as you want in Canvas prior to the deadline. Apparently, if you resubmit it after the deadline, one (or more) tokens will be used. In that case, also let me know via e-mail at the time of the deadline that you are still working on your HW beyond the original deadline and plan to resubmit a new version later on. Once a HW has been graded on my side, you will no longer be allowed to resubmit a revision of that HW (or in case you did, you still would get the points based on the last submission prior to the original deadline).

### **Project (Stat 6080 only):**

There will be one major project towards the end of the semester. This will include the preparation of a final project report and a short presentation of your work for the other students in this course. The project will be done in a small group of students. **The project will account for 30% of your course grade.**

### **Grading Scheme:**

I do not specify any definite grade cutoffs in advance, but I typically assign the same grade to all students that fall into the same narrow cluster of points that extends below one of the commonly used grade cutoffs (i.e., 93% for A, 90% for A–, 87% for B+, 83% for B, 80% for B–, 77% for C+, 73% for C, 70% for C–, 67% for D+, 60% for D, and <60% for F). So, for example, if one student earned 92.9% of points and the next higher student

earned 93.1% of points, both students almost certainly will get an A grade. However, if one student earned 92.9% of points and the next higher student earned 95.0% of points, this will almost certainly translate into A– and A grades, respectively.

### Textbooks:

Murrell, Paul (2009) *Introduction to Data Technologies*, Boca Raton, FL: Chapman and Hall/CRC,

<https://www.routledge.com/Introduction-to-Data-Technologies/Murrell/p/book/9781138118027>, and

<https://www.stat.auckland.ac.nz/~paul/ItDT/>.

Nolan, Deborah, and Temple Lang, Duncan (2015) *Data Science in R — A Case Studies Approach to Computational Reasoning and Problem Solving*, Boca Raton, FL: CRC Press/Taylor & Francis,

<https://www.routledge.com/Data-Science-in-R-A-Case-Studies-Approach-to-Computational-Reasoning-and-Problem-Solving/Nolan-Lang/p/book/9781482234817>.

Every student should have access to each of these books, but it is not necessary that every student buys all of these books. The USU library holds several of these books or provides online access. If you plan to work in the area of Data Science for your MS or PhD degree, you should consider to purchase these books for an ongoing use beyond this course.

### Software:

We will primarily be using R (<https://cran.r-project.org/>), a free software environment for statistical computing and graphics. Please install a recent version of R, i.e., 4.6.1 or later, on your own computer so we can exchange code. Working with version 4.6.0 should still be OK. I would discourage the use of version 4.5.3 or older. Also install the most recent version (2026.08.2 Build 200, released on 2026-08-26) of RStudio (<https://posit.co/>) as a front end to R. **Note that some versions of RStudio released earlier in 2024 contained bugs that prevented them from opening certain Rnw files, similar to the ones used in class. If you get error messages when you try to open some of the files from class, check whether you have the most recent version of RStudio installed — and if not, upgrade to the most recent version.** For Windows, also install Rtools 4.5 (<https://cran.r-project.org/bin/windows/Rtools/>).

For most work in this course, using the *tinytex* R package is all that is needed, but if you plan to work on more complicated L<sup>A</sup>T<sub>E</sub>X documents later on (such as your MS thesis or your PhD dissertation), you should also install a full version of L<sup>A</sup>T<sub>E</sub>X that can be used in connection with RStudio. My personal recommendation is TeX Live (<https://www.tug.org/texlive/>).

### Credits:

This course uses some of the course materials provided by Dr. Paul Murrell (University of Auckland: <https://www.stat.auckland.ac.nz/~paul/>), Dr. Duncan Temple Lang (UC Davis: <https://www.stat.ucdavis.edu/~duncan/>) and Dr. Deborah Nolan (UC Berkeley: <https://www.stat.berkeley.edu/~nolan/>). We are likely to include parts from additional web sources that will be specified later on.

### Courtesy:

Please turn off cell phones and similar devices before class, and please keep conversations to a minimum during lectures. Please do not read/reply to your e-mails or browse other web pages than the ones discussed during class.

I will not keep track if you come to class or not. However, I would highly recommend to attend all lectures so you can immediately ask questions and earn LAPs when assigned.

**Americans with Disabilities Act:**

USU welcomes students with disabilities. If you have, or suspect you may have, a physical, mental health, or learning disability that may require accommodations in this course, please contact the Disability Resource Center (DRC) as early in the semester as possible (University Inn Room 101, 435-797-2444, [drc@usu.edu](mailto:drc@usu.edu), <https://www.usu.edu/drc/>). All disability related accommodations must be approved by the DRC. Once approved, the DRC will coordinate with faculty to provide accommodations.

**Note:**

The above schedule and procedures in this course are subject to change in the event of extenuating circumstances.